Aletheia

AI Business Knowledge Assistant

Ask a business question in plain language. It routes to the right source, answers with hybrid SQL + document retrieval, and attaches a citation to every fact — one you can trace to the exact database row or document page.

▶ Try the live demo · **View the source**

`query routing · hybrid SQL + RAG · grounded generation · source attribution · English + Hebrew`

> **This is not a PDF chatbot**, and not "upload PDFs into a vector database and do semantic search" — the approach you explicitly ruled out. It is a real **query-routing → hybrid-retrieval → grounded-generation → source-attribution** pipeline, built to extend to your future CRM / email / cloud-storage / case-management sources.

At a glance

What it does	Routes each question to the relevant source(s), retrieves with SQL + RAG, returns a cited answer
The differentiator	Honest grounding — every claim cites a resolvable source, or it says the data doesn't have it
Capabilities	Contract Intelligence · Case File Q&A · Maintenance Spend Intelligence
Languages	English + Hebrew (live today)
Stack	Next.js · DeepSeek · local multilingual embeddings · bundled SQLite + vector index · Vercel
Quality floor	40 unit + 13 journey tests (run against the live deploy) · CI-enforced

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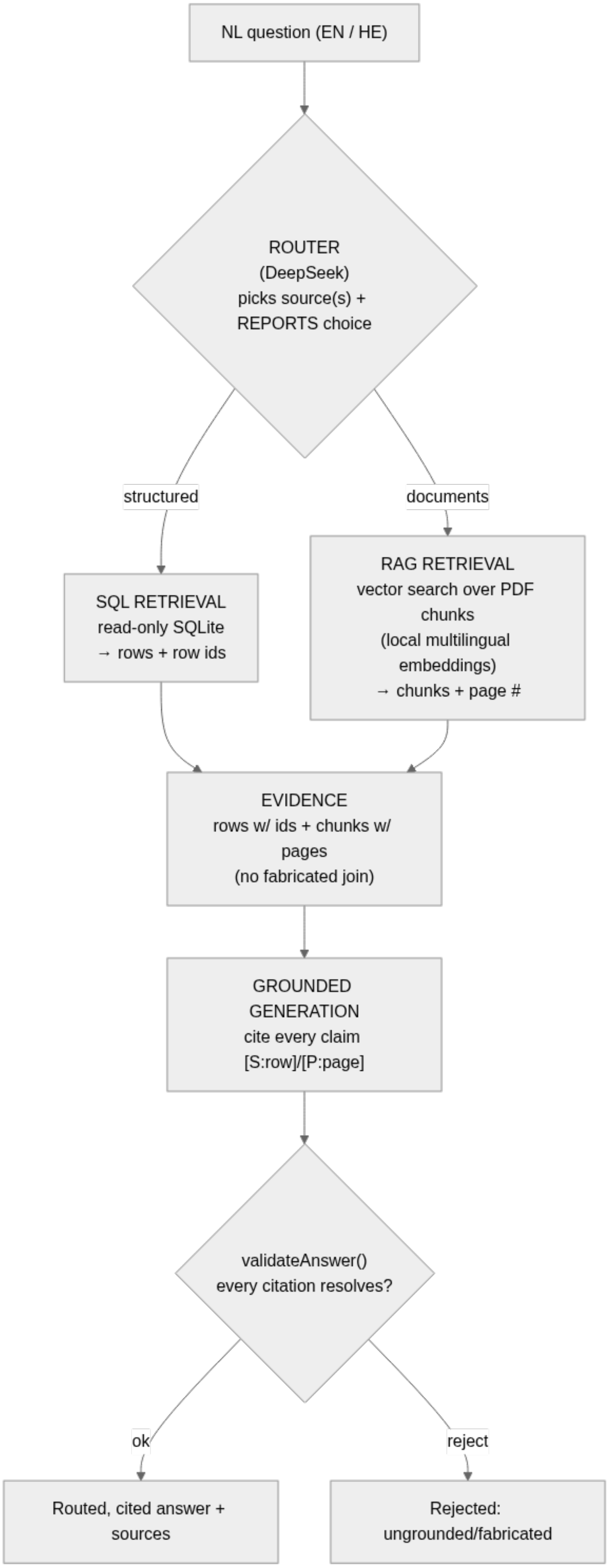
How it works



Architecture: routing → hybrid retrieval → grounded generation → validation

1. **Query routing** — a real LLM decision picks the relevant source(s) and **reports it** in the UI (it's not a hidden heuristic).
2. **Hybrid retrieval** — SQL over your structured data (every row carries a stable id) + RAG over your documents (every chunk carries its page).
3. **Grounded generation** — the answer is composed **only** from retrieved evidence, with an inline citation on every fact.
4. **Source attribution + a content-fidelity gate** — `validateAnswer()` rejects any answer whose citation doesn't resolve to real evidence, so "fluent but ungrounded" can't ship. The two source domains (structured vs. documents) are **never merged on an invented join**.

Full architecture + the data-flow diagram: architecture.md. Engine internals: features/shared-engine/reference.md.

What it can do (three worked capabilities)

Each capability is a real feature with its own golden examples, regression tests, and the screenshots below — all captured from the live demo.

1. Contract Intelligence — expiry, value, and honest about penalties

Ask "What contracts expire in the next 90 days and what penalties are defined in those contracts?" → **38 contracts** expiring, combined annual value **$18,924,883.79**, each row cited — and an **honest statement that penalty terms are not available** (your contract data has no penalty field and no contract documents). It never fabricates a penalty and never pulls from the unrelated case file.

Contract Intelligence — 38 expiring, $18.9M, cited rows, honest on penalties

Full guide: contract-intelligence/user-guide.md.

2. Case File Q&A — page-cited findings, corroboration, and surfaced conflicts

Ask "What was the final child support amount, and who got primary residence?" → **$1,285/month** + **primary residence to Joni Carter**, cited to **Page 24 (Final Judgment)**. Ask about the grounds → **corroborated across both documents**. Ask the filing date → it **surfaces the conflict** (the cover sheet says 10 Feb, the narratives say 3 Feb) instead of silently picking one.

Case File Q&A — the filing-date conflict surfaced with both citations

Full guide: case-file-qa/user-guide.md.

3. Maintenance Spend Intelligence — cited spend, and honest refusal over fiction

Ask "How much did we spend on maintenance in 2026, and which vendors cost the most?" → **$13,485.66 across 248 tickets** (2026), **$40,597.00 all-time**, top vendors cited and drillable. Ask the literal "which customers have overdue payments and what does the agreement say about suspension?" → it **honestly refuses**: this data has no payment-status field and no service agreement, so it explains why and **pivots to the spend analysis it can do** — rather than inventing an overdue list.

Aletheia.

KNOWLEDGE ASSISTANT

Ask a business question in plain language. Aletheia **routes** *it to the right source, answers with* **hybrid SQL + document retrieval***, and attaches a* **citation to every fact** *you can trace to the exact row or page. Not a PDF chatbot — grounded, or it says so.*

❧ Which customers have overdue payments and what does the agreement say about service suspe ASK

ROUTING DECISION

question → ● Structured · SQL

The question asks about overdue payments and service suspension terms, which are business operations questions; the structured maintenance_spend intent will honestly report that the data has no such fields, and the documents contain no business agreement.

intents: maintenance_spend

ANSWER

The maintenance data contains no payment-status, paid/unpaid, due-date, or service-suspension field, and there is no service-agreement document. The columns that exist are: Ticket ID, Vendor, Invoice, Labor Cost, Parts Cost, Total Cost, and Completion Date. The vendors are maintenance providers the school pays — they are not customers who owe money. Therefore, I cannot identify which customers have overdue payments or state what the agreement says about service suspension from this data.

However, the total maintenance spend is $40,597.00 across 750 tickets. I can break this down by vendor or year if needed.

Verified figures (computed over the cited maintenance rows):
- total maintenance spend is $40,597.00 across 750 tickets [S:maintenance#5].
- top vendor by spend is Oyoba at $949.94 across 3 tickets [S:maintenance#5].

✓ Grounded — every cited source resolves to retrieved evidence (validateAnswer passed).

SOURCES — 15 ROWS · 0 DOCUMENT CHUNKS

[S:maintenance#5] MAINTENANCE · ROW 5
ticket_id: Cookies · vendor: Oyoba · total_cost: 549.98 · completion_date: 3/7/2024

[S:maintenance#685] MAINTENANCE · ROW 685
ticket_id: Juice · vendor: Zoovu · total_cost: 499.98 · completion_date: 10/31/2026

[S:maintenance#257] MAINTENANCE · ROW 257
ticket_id: Jigsaw Puzzles · vendor: Voolith · total_cost: 412.98 · completion_date: 8/18/2024

[S:maintenance#234] MAINTENANCE · ROW 234
ticket_id: Specialty Salts · vendor: Kayveo · total_cost: 399.98 · completion_date: 2/7/2026

[S:maintenance#471] MAINTENANCE · ROW 471
ticket_id: Poultry · vendor: Dabtype · total_cost: 389.98 · completion_date: 2/8/2025

▾ Show all 15 rows

← Ask another question

Aletheia · grounded knowledge assistant · source · routing · hybrid SQL + RAG · cited & verified

Maintenance — honest refusal of the overdue question, pivoting to real spend

Full guide: maintenance-spend-intelligence/user-guide.md.

The trust property (why this isn't naive RAG)

The single thing that distinguishes a real knowledge assistant from "PDFs in a vector DB" is that it is **honest about what it doesn't know**. This MVP makes that a tested guarantee, not a hope:

- **Every factual claim cites a resolvable source** (a SQLite row id or a PDF page). An answer that makes an uncited claim is **rejected** by `validateAnswer()`.
- **It states absence instead of fabricating** — no penalty data → "not available"; no payment-status field → "I can't determine overdue payments, and here's why"; a source conflict → both values surfaced, not silently resolved.
- **It never invents a join** between unrelated sources (your school operations data and the Carter case file share no key — they're composed and cited separately, never merged).

These behaviors are enforced as **red automated tests** (per-feature content-fidelity gates + journey tests that run against the live deployment), not described in prose.

Data Quality Assessment

> **This is a selling point, not a footnote.** Your stated fear is naive work — "someone who simply uploads PDFs into a vector database and performs semantic search." The antidote is **source-vetting judgment**: knowing what your data can and cannot honestly answer before building. We inspected **all 9 provided sources at the row level**, built features only where a source can ground a real, cited answer, and **deliberately dropped the five that can't** — naming the exact defect for each.

4 of 9 sources support real, grounded features. 5 were vetted and dropped — for specific, named defects, not for time.

#	Source	Verdict	The defect that decides it
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1	`school data 1.csv` — contracts	✓ **Built on**	Clean costs/dates → Contract Intelligence. (`Contract ID` actually holds job titles; some End<Start; no penalty column — handled honestly.)
2	`school data 3.csv` — maintenance	✓ **Built on**	Clean spend → Maintenance Spend. No payment-status/due-date field → overdue questions honestly refused, not faked.
3	Family Court Case File (PDF)	✓ **Built on**	Page-numbered, citable → Case File Q&A (Page-24 Final Judgment).
4	Carter Story (PDF)	✓ **Built on**	Corroborating narrative for the case file.
5	`school data 2.csv` — enrollment	✗ **Dropped**	`term_name` is 100% a Ruby error string; `status` holds gender values — columns are the **wrong data entirely**.

#	Source	Verdict	The defect that decides it
6	`school data 4.csv` — payroll	⚠ **Deferred**	Pay-method/notes columns are 100% error strings; the headline fields are corrupt.
7	`school data 6.csv` — payroll (alt)	⚠ **Deferred**	`pay_month` ranges 1–100; payment method is a random integer; can't reconcile with #6.
8	`school data 5.csv` — invoice totals	✗ **Dropped**	**All 788 rows are identical** (180/6/1080) — zero variance to analyze or cite.
9	`school data .csv` — person list	✗ **Out of scope**	Generic directory (incl. PII-shaped fields); no spec question touches it.

That discipline is the product — the system says "I can't answer that from this data" (the maintenance-overdue and contract-penalty cases) rather than fabricate, applied not just at answer time but at **data-intake time**.

Full row-level evidence — docs/product/data-quality-assessment.md (every source, the exact defect, and what clean re-supply would un-block).

Extensible by design

The router + retrieval layer is **source-pluggable**. Your future integrations — CRM, email, cloud storage, case management — each register as a new retrieval source behind the same router contract, **without changing** the grounding/citation/validation layer. The per-fact citation model is what makes per-source access control addable later.

English + Hebrew

The demo answers in **both English and Hebrew today** (the embedding model is multilingual; citations are preserved across languages). Each feature's guide includes a Hebrew golden example. RTL UI polish and Hebrew-tuned prompts are the natural next step — the architecture is already ready for them.

Stack

Next.js · DeepSeek (`deepseek-chat`) for routing + generation · local `Xenova/multilingual-e5-small` embeddings (no embeddings key; the Hebrew seam) · bundled read-only SQLite + an in-app vector index · deployed on Vercel. Self-contained and reproducible (`npm run build:index` rebuilds the index deterministically from the source data).

Quality floor: every push runs CI — doc-lint + doc-structure-lint + typecheck + 40 unit tests; 13 journey tests run against the live deployment. The content-fidelity gates above are part of that floor.